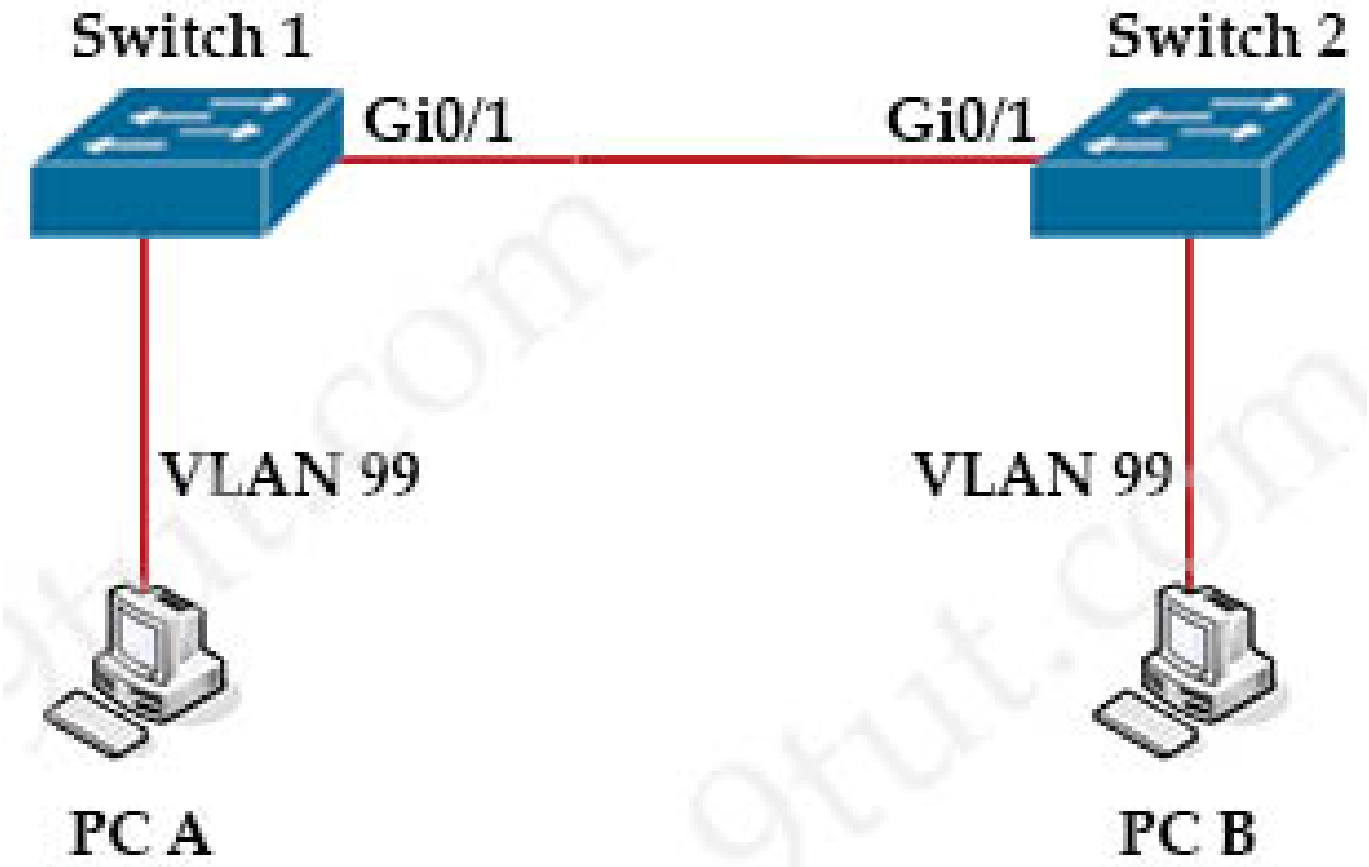
VLAN & Trunking Questions

10 questions | Original English questions and source answers

Question 1

Refer to the exhibit. After the switch configuration the ping test fails between PC A and PC B. Based on the output for switch 1, which error must be corrected?
Switch 1
Name: Gi0/1
Switchport: Enabled
Administrative Mode: trunk
Operational Mode: trunk
Administrative Trunking Encapsulation: dot1q
Operational Trunking Encapsulation: dot1q
Negotiation of Trunking: Off
Access Mode VLAN: 1 (default)
Trunking Native Mode VLAN: 1 (default)
Switch 2
Name: Gi0/1
Switchport: Enabled
Administrative Mode: trunk
Operational Mode: trunk
Administrative Trunking Encapsulation: dot1q
Operational Trunking Encapsulation: dot1q
Negotiation of Trunking: Off
Access Mode VLAN: 1 (default)
Trunking Native Mode VLAN: 99 (default)
Administrative Native VLAN tagging: enabled
Voice VLAN: none
<output omitted>
Trunking VLANs Enabled: 50-100
Pruning VLANs Enabled: 2-1001
Capture Mode Disabled
Capture VLANs Allowed: ALL
Administrative Native VLAN tagging: enabled
Voice VLAN: none
<output omitted>
Trunking VLANs Enabled: 50-100
Pruning VLANs Enabled: 2-1001
Capture Mode Disabled
Capture VLANs Allowed: ALL



Options

A. There is a native VLAN mismatch

B. Access mode is configured on the switch ports.

C. The PCs are in the incorrect VLAN

D. All VLANs are not enabled on the trunk

Answer

A. There is a native VLAN mismatch

Source explanation

From the output we see the native VLAN of Switch1 on Gi0/1 interface is VLAN 1 while that of Switch2 is VLAN 99 so there would be a native VLAN mismatch.

Source: CCNA Training » Trunking Quiz.pdf | Question 1

Question 2

Two switches are connected and using Cisco Dynamic Trunking Protocol. SW1 is set to Dynamic Desirable. What is the result of this configuration?

Options

A. The link is in a downstate.

B. The link is in an error disables stale

C. The link is becomes an access port

D. The link becomes a trunkport

Answer

D. The link becomes a trunkport

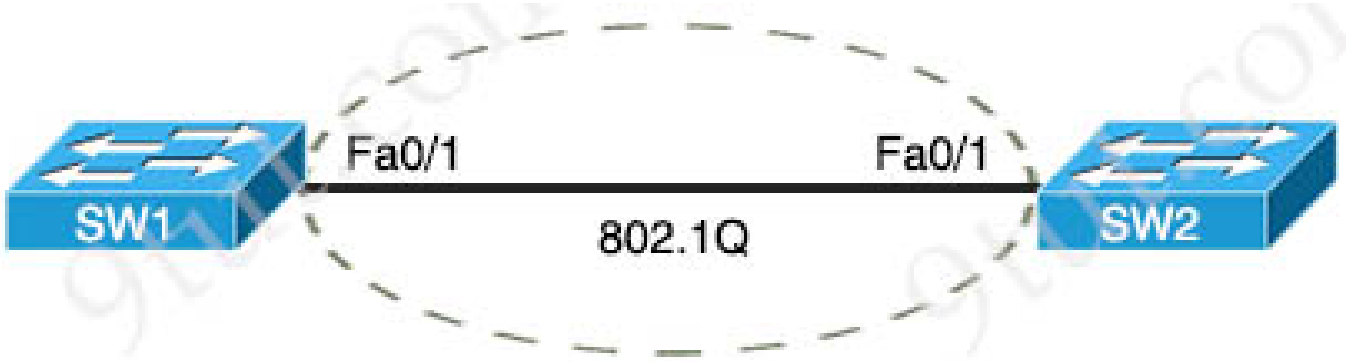
Source explanation

Maybe this question is missing the “SW2 is set to Dynamic Auto” part so we assume this part to find out the best answer. Dynamic Desirable + Dynamic Desirable/Dynamic Auto/Trunk will form a trunk link.

Source: CCNA Training » Trunking Quiz.pdf | Question 2

Question 3

Refer to exhibit. Which action do the switches take on the trunk link?
interface FastEthernet0/1
switchport trunk encapsulation dot1q
switchport trunk native vlan 999
switchport mode trunk
interface FastEthernet0/1
switchport trunk encapsulation dot1q
switchport trunk native vlan 99
switchport mode trunk



Options

A. The trunk does not form and the ports go into an err-disabled status

B. The trunk forms but the mismatched native VLANs are merged into a single broadcast domain

C. The trunk does not form, but VLAN 99 and VLAN 999 are allowed to traverse the link

D. The trunk forms but VLAN 99 and VLAN 999 are in a shutdown state

Answer

B. The trunk forms but the mismatched native VLANs are merged into a single broadcast domain

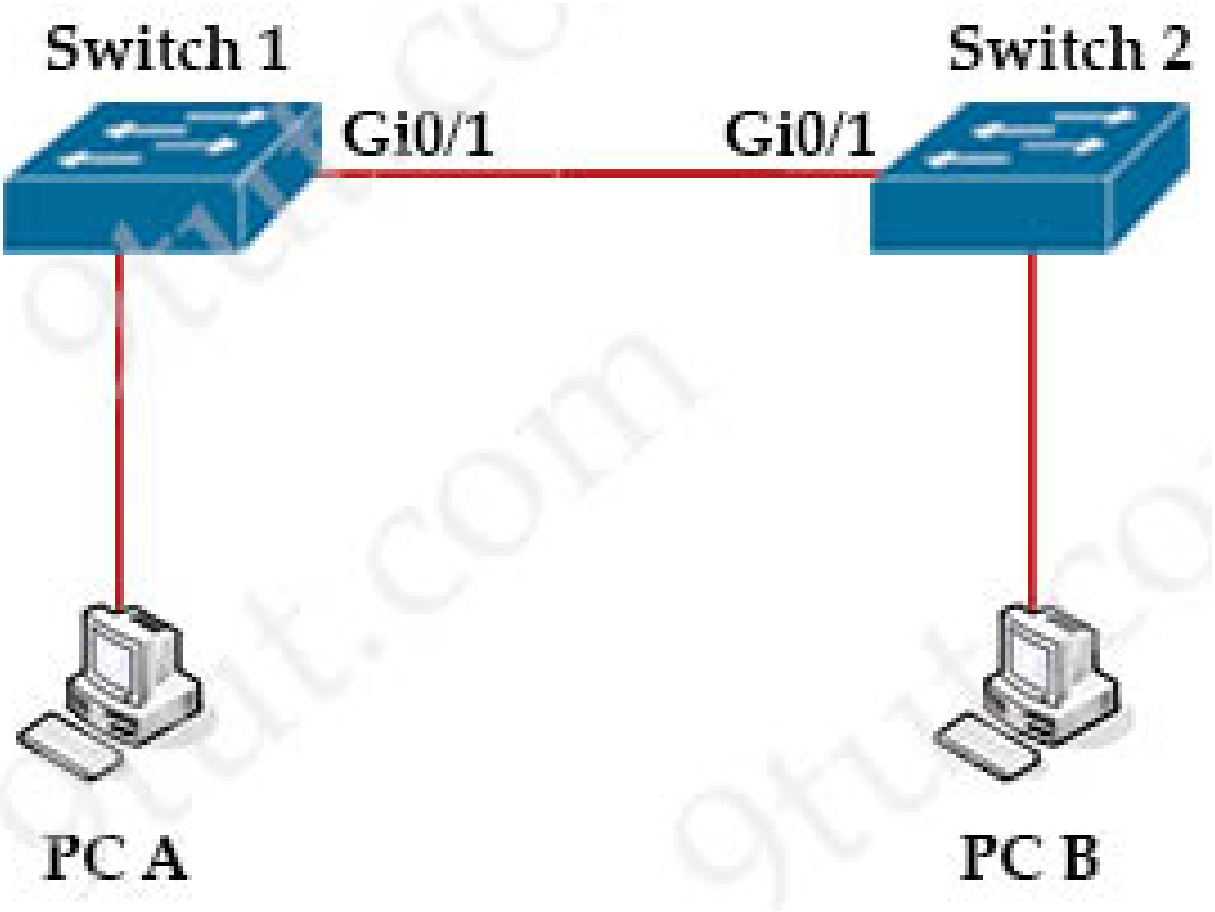
Source explanation

The trunk still forms with mismatched native VLANs and the traffic can actually flow between mismatched switches. But it is absolutely necessary that the native VLANs on both ends of a trunk link match; otherwise a native VLAN mismatch occurs, causing the two VLANs to effectively merge. For example with the above configuration, SW1 would send untagged frames for VLAN 999. SW2 receives them but would think they are for VLAN 99 so we can say these two VLANs are merged.

Source: CCNA Training » Trunking Quiz.pdf | Question 3

Question 4

Refer to the exhibit. The network administrator wants VLAN 67 traffic to be untagged between Switch 1 and Switch 2 while all other VLANs are to remain tagged. Which command accomplishes this task?



Options

A. `switchport access vlan 67`

B. `switchport trunk allowed vlan 67`

C. `switchport private-vlan association host 67`

D. `switchport trunk native vlan 67`

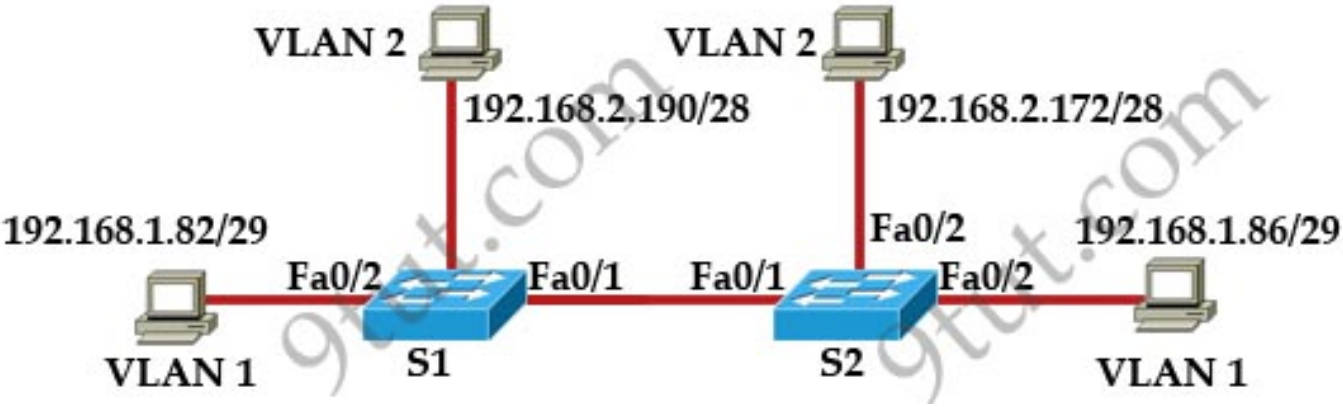
Answer

D. switchport trunk native vlan 67

Source: CCNA Training » Trunking Quiz.pdf | Question 4

Question 5

Refer to the exhibit.
A frame from VLAN1 of switch S1 is sent to switch S2 where the frame received on VLAN2. What causes this behavior?




```
S1#show interface trunk
Port        Mode         Encapsulation  Status        Native vlan
Fa0/1       on           802. 1q        Trunking      1

Port        Vlans allowed a trunk
Fa0/1       1.1005

Port        Vlans allowed and active in management domain
Fa0/1       12
```

```
S2#show interface trunk
Port        Mode         Encapsulation  Status        Native vlan
Fa0/1       on           802. 1q        Trunking      2

Port        Vlans allowed a trunk
Fa0/1       1.1005

Port        Vlans allowed and active in management domain
Fa0/1       12
```

Options

A. trunk mode mismatches

B. VLANs that do not correspond to a unique IP subnet

C. native vlan mismatches

D. allowing only vlan 2 on the destination

Answer

C. native vlan mismatches

Source explanation

For 802.1q encapsulation, the native VLAN must matched at both side; otherwise the link will not work. In this case the native VLAN of S1 is 1 while the native VLAN of S2 is 2. Therefore when a frame from VLAN 1 is sent from S1 to S2, it will be untagged before leaving S1. When S2 receives this untagged frame, it believes this
frame was sent from the native VLAN. But the native VLAN on S2 is VLAN 2, not VLAN 1. Therefore only hosts on VLAN2 of S2 will receive this frame.

Source: CCNA Training » Trunking Quiz.pdf | Question 5

Question 6

Refer to the exhibit. Which action is expected from SW1 when the untagged frame is received on the GigabitEthernet0/1 interface?

```
SW1#show run int gig 0/1
interface GigabitEthernet0/1
switchport access vlan 11
switchport trunk allowed vlan 1-10
switchport trunk encapsulation dot1q
switchport trunk native vlan 5
switchport mode trunk
speed 1000
duplex full
```

Options

A. The frame is processed in VLAN 5

B. The frame is processed in VLAN 11

C. The frame is processed in VLAN 1

D. The frame is dropped

Answer

A. The frame is processed in VLAN 5

Source: CCNA Training » Trunking Quiz.pdf | Question 6

Question 7

Refer to the exhibit.
A network administrator is configuring an EtherChannel
between SW1 and SW2. The SW1 configuration is shown. What is the correct configuration for SW2?



SW1

```
interface FastEthernet 0/1
 channel-group 1 mode auto
 switchport trunk encapsulation dot1q
 switchport mode trunk

interface FastEthernet 0/2
 channel-group 1 mode auto
 switchport trunk encapsulation dot1q
 switchport mode trunk
```

Options

```
A. interface FastEthernet 0/1
channel-group 1 mode active
switchport trunk encapsulation dot1q
switchport mode trunk
!
interface FastEthernet 0/2
channel-group 1 mode active
switchport trunk encapsulation dot1q
switchport mode trunk
```

```
B. interface FastEthernet 0/1
channel-group 2 mode auto
switchport trunk encapsulation dot1q
switchport mode trunk
!
interface FastEthernet 0/2
channel-group 2 mode auto
switchport trunk encapsulation dot1q
switchport mode trunk
```

```
C. interface FastEthernet 0/1
channel-group 1 mode desirable
switchport trunk encapsulation dot1q
switchport mode trunk
!
interface FastEthernet 0/2
```

```
channel-group 1 mode desirable
switchport trunk encapsulation dot1q
switchport mode trunk
```

```
D. interface FastEthernet 0/1
channel-group 1 mode passive
switchport trunk encapsulation dot1q
switchport mode trunk
!
interface FastEthernet 0/2
channel-group 1 mode passive
switchport trunk encapsulation dot1q
switchport mode trunk
```

Answer

C. interface FastEthernet 0/1
channel-group 1 mode desirable
switchport trunk encapsulation dot1q
switchport mode trunk
!
interface FastEthernet 0/2
channel-group 1 mode desirable
switchport trunk encapsulation dot1q
switchport mode trunk

Source explanation

From the configuration of SW1, we see it is using Port Aggregation Protocol (PAgP) with “auto” mode so the other end (SW2) must also runs PAgP with “desirable” mode to actively sending request to form an Etherchannel.

Source: CCNA Training » Trunking Quiz.pdf | Question 7

Question 8

What occurs to frames during the process of frame flooding?

Options

A. Frames are sent to all ports, including those that are assigned to other VLANs

B. Frames are sent to every port on the switch that has a matching entry in MAC address table

C. Frames are sent to every port on the switch in the same VLAN except from the originating port

D. Frames are sent to every port on the switch in the same VLAN

Answer

C. Frames are sent to every port on the switch in the same VLAN except from the originating port

Source: CCNA Training » Trunking Quiz.pdf | Question 8

Question 9

Refer to the exhibit. An administrator is tasked with configuring a voice VLAN. What is the expected outcome when a Cisco phone is connected to the GigabitEthernet 3/1/4 port on a switch?
interface GigabitEthernet3/1/4
switchport voice vlan 50
!

Options

A. The phone and a workstation that is connected to the phone do not have VLAN connectivity.

B. The phone sends and receives data in VLAN 50, but a workstation connected to the phone sends and receives data in VLAN 1.

C. The phone sends and receives data in VLAN 50, but a workstation connected to the phone has no VLAN connected.

D. The phone and a workstation that is connected to the phone send and receive data in VLAN 50.

Answer

B. The phone sends and receives data in VLAN 50, but a workstation connected to the phone sends and receives data in VLAN 1.

Source: CCNA Training » Trunking Quiz.pdf | Question 9

Question 10

Refer to the exhibit. What commands are needed to add a subinterface to Ethernet0/0 on R1 to allow for VLAN 20, with IP address 10.20.20.1/24?
R1
interface Ethernet0/0
no ip address
SW2:
interface Ethernet0/1
switchport trunk encapsulation dot1q
switchport mode trunk
!
interface Ethernet0/2
switchport access vlan 20
switchport mode access
SW1:
interface Ethernet0/0
switchport trunk encapsulation dot1q
switchport mode trunk
!
interface Ethernet0/1
switchport trunk allowed vlan 10
switchport trunk encapsulation dot1q
switchport mode trunk



Options

```
A. R1 (config)#interface ethernet0/0
R1 (config)#encapsulation dot1q 20
R1(config)#ip address 10.20.20.1 255.255.255.0

B. R1 (config)#interface ethernet0/0.20
R1 (config)#encapsulation dot1q 20
R1(config)#ip address 10.20.20.1 255.255.255.0

C. R1 (config)#interface ethernet0/0.20
R1(config)#ip address 10.20.20.1 255.255.255.0

D. R1 (config)#interface ethernet0/0
R1(config)#ip address 10.20.20.1 255.255.255.0
```

Answer

B. R1 (config)#interface ethernet0/0.20
R1 (config)#encapsulation dot1q 20
R1(config)#ip address 10.20.20.1 255.255.255.0

Source: CCNA Training » Trunking Quiz.pdf | Question 10